Partial Price Transparency and Demand Learning in Retail Price Competition

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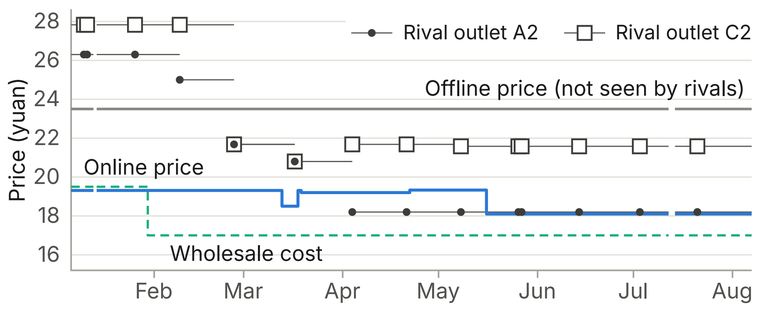
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Full paper
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1 One product in one trade area

A pharmacy chain sells a 15 g tube of desonide cream 0.05% in its store and through a delivery platform. Its staff see rivals' online prices, not their costs or store prices, and must decide which rival listing is the same product. After the chain's wholesale cost fell in late January, both matched rivals' recorded prices fell. By mid-May the chain had cut its online price just below the cheaper of the two. Its store price never changed.

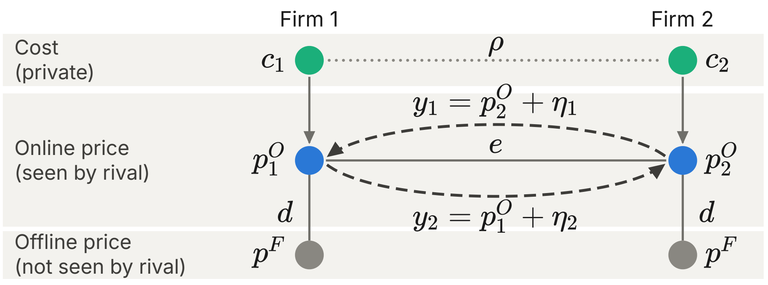


Step lines are the chain's prices and wholesale cost; markers are the two rivals' listed online prices on the days they were recorded, and thin lines carry the last record forward (the records coincide on 26 February and 17 March). The chain changed its online price five times, four times to below the cheapest rival record up to that day.

We call the rival price a seller records its **quote**, the gap to the rival's actual price the **wedge**, and the inverse variance of the wedge the quote's **precision**. A seller that estimates demand from its own sales must control for the rival's price, and it can do so only through its quote.

Question Does higher precision make prices more competitive, as retail gasoline suggests (Byrne et al. 2025, *JPE*)?

2 The model



Arrows from costs: pass-through; solid lines: substitution in demand (d within a firm, e across firms); dotted line: cost correlation; dashed arrows: the quotes, with wedges η.

- Two symmetric firms sell substitutes. Costs are private with correlation ρ; each firm's offline store price is fixed and not seen by the rival.
- Firm i sees the rival's online price only through its quote $y_i = p_{-i}^O + \eta_i$, with wedge $\eta_i \sim \mathcal{N}(0, \sigma_\eta^2)$ and precision $1/\sigma_\eta^2$.
- Online demand is $q_i^O = \alpha - b\,p_i^O + d\,p_i^F + e\,p_{-i}^O + \epsilon_i^O$, with intensity of competition $\omega = e/(2b)$.

Firms know their own two channels but not the online demand parameters (α, b, e). Each period an adaptive firm prices, observes its sales and adapts in one of two ways; a fixed rule is the third benchmark.

- **Model-based (M)**
  Regresses its online sales on its own price and its quote, and prices against the estimate until the data its prices generate return the same estimate (a self-confirming point, Fudenberg and Levine 1993).
- **Rule-tuning (R)**
  Holds a fixed belief $\tilde{b}$ about its own price sensitivity and moves its pricing rule in the direction of its marginal profit, taking the rival's price as given.
- **Fixed rule (F)**
  Never revises its rule. The chain prices this way, and by its management's account its rivals price by judgment.

Benchmark (paper Table 2): $\alpha = 30$, $b = 2$, $e = 1$ ($\omega = 0.25$), $d = 0.5$, $\bar{c} = 8$, $\sigma_c^2 = 4$, $\rho = 0.3$, commission $\phi = 0.15$, $p^F = 25$; Nash mean price 23.77. Simulated costs are truncated at 3 standard deviations, giving $\sigma_c^2 = 3.885$ and $\rho = 0.293$; results at $\rho = 0.293$ use these moments.

3 Where the regression bias comes from

A model-based seller regresses its sales on its own price and its quote y_i. The part of the rival's price that the quote misses moves with the seller's own price in two ways: through the cost correlation (ρ) and through the rival's response to its quote of the seller's price (its rule's weight π_x on the quote). Least squares attributes this co-movement to the own price.

When both firms use the same affine rule ($\pi_c > 0$, $\pi_x \in [0, 1)$), the population least-squares slope on the own price is

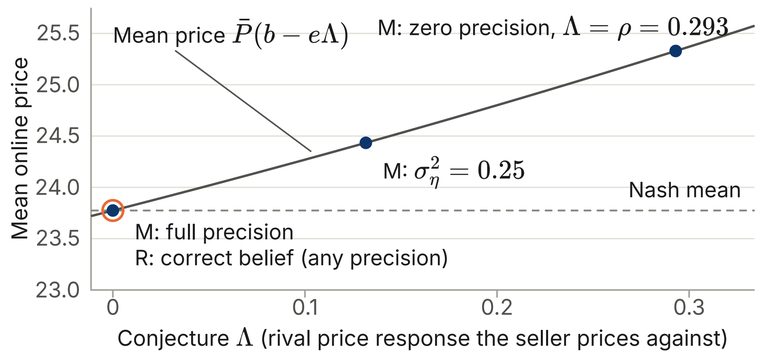
$$\hat{b} = b - \frac{e\,\kappa\,(\pi_x + \rho)}{1 - \rho^2 + \kappa\,(1 + \pi_x^2)}, \qquad \kappa = \frac{\sigma_\eta^2}{\pi_c^2 \sigma_c^2},$$

where π_c is the rule's weight on own cost and κ is the wedge variance relative to the cost-driven price variance. At zero precision $\pi_x = 0$ and $\kappa \to \infty$, so $\hat{b} = b - e\rho$; at full precision $\kappa = 0$ and $\hat{b} = b$. With unequal rules only the rival's weight on its quote enters the bias (paper Lemma 2).

In the panel of Section 9 the online lag sum is 0.50. Read as a rival's weight π_x on the chain's price, it means a demand model without rival prices would understate the chain's price sensitivity by nearly half at the benchmark (paper Section 6.3).

4 Stopping points as conjectural variations

At every stopping point of both adaptive families, the mean price equals that of a conjectural-variations equilibrium (Bresnahan 1981; Light and Wang 2026): a seller prices as if the rival's online price moved by Λ per unit change in its own. The mean price depends only on Λ, equals $\bar{P}(b - e\Lambda)$ and rises with Λ, so precision can move the mean price only by moving Λ.



The mean price $\bar{P}(b - e\Lambda)$ against Λ at the benchmark, with the stopping points of the two families (Family M at the truncated moments, $\rho = 0.293$). A rule-tuning seller with belief $\tilde{b} = 1.7$ stops at $\Lambda = 0.30$ with mean price 25.37, next to zero-precision Family M ($\Lambda = 0.293$, 25.33), and is not drawn, to keep the markers apart.

- **Rule-tuning:** $\Lambda = (b - \tilde{b})/e$ is set by the belief, so precision does not move the mean price; with a correct belief $\Lambda = 0$ and precision only makes the two prices move together.
- **Model-based:** $\Lambda = (b - \hat{b})/e$ is set by the regression, with $\Lambda = \rho$ at zero precision and $\Lambda = 0$ at full precision. Underestimating the own price sensitivity ($\hat{b} < b$) means a mean price above the Nash mean.

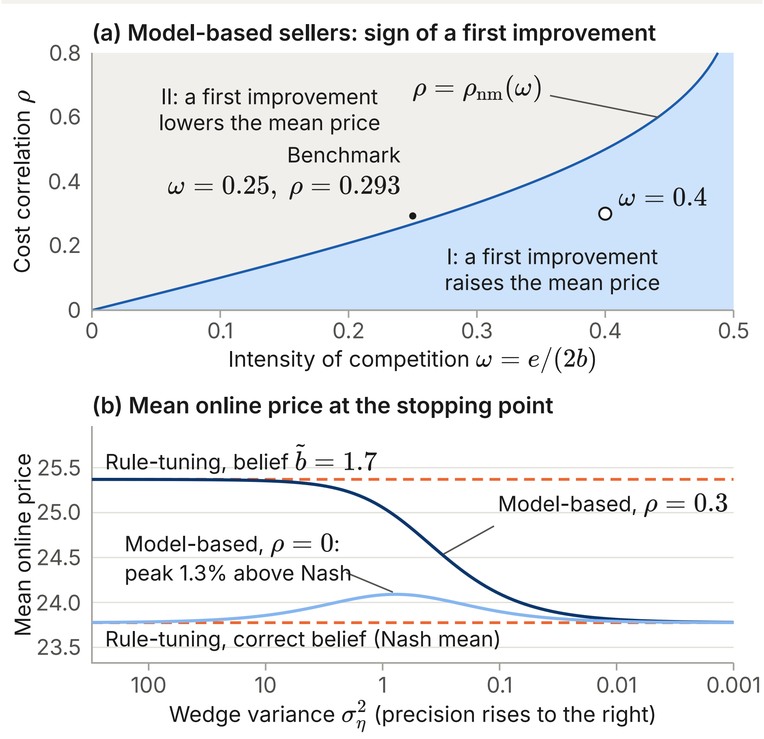
5 Partial precision: two opposing effects

- **Response effect:** the rival starts to respond to its quote of the seller's price, the co-movement grows, $\hat{b}$ falls and the price rises.
- **Control effect:** the regression controls for more of the rival's price; $\hat{b}$ rises and the price falls.

Theorem 2 When the self-confirming point is unique, a first improvement from zero precision raises a model-based seller's mean price if and only if

$$\rho < \rho_{\mathrm{nm}}(\omega) = \frac{1 - \sqrt{1 - 4\omega^2}}{2\omega} \qquad \left(\omega < \tfrac{1}{2}\right),$$

where $\rho_{\mathrm{nm}}(\omega)$ is the consistent conjecture of Bresnahan (1981). Near full precision the bias falls with precision for every ρ.



(a) Region I lies below $\rho_{\mathrm{nm}}(\omega)$; the benchmark $\rho = 0.293$ is just above $\rho_{\mathrm{nm}}(0.25) = 0.268$; the open circle is $\omega = 0.4$, $\rho = 0.3$. (b) Benchmark with $\sigma_c^2 = 4$: the model-based price falls with precision at $\rho = 0.3$ and rises before it falls at $\rho = 0$; the rule-tuning price does not move. With more intense competition and no cross-channel substitution ($\omega = 0.4$, $d = 0$), the peak at $\rho = 0$ is 4.2% above the Nash mean.

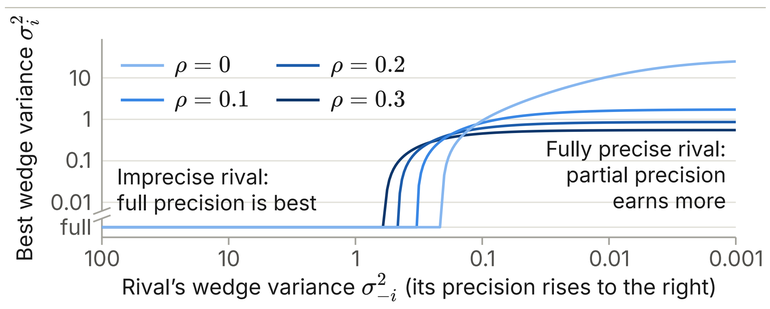
6 Welfare depends on how sellers adapt

At the benchmark ($\rho = 0.293$), from zero to full precision:

Seller	Mean price	Consumer surplus
● Model-based (M)	−6.1%	+6.6%
○ Rule-tuning, correct belief (R)	unchanged	−0.013%

- Under the condition of Proposition 3, met at the benchmark, consumer surplus under Family M rises from zero to full precision if and only if $\rho > \rho^\star$, and $\rho^\star = 0.00065$ at the benchmark ($\sigma_c^2 = 4$). At $\rho = 0.293$ (truncated moments) the consumer gain is more than 400 times the loss under rule-tuning with a correct belief.
- Under rule-tuning with a correct belief ($\omega \le 1/2$), precision only synchronizes prices: the platform's commission revenue rises, and total surplus falls at every commission up to 1/5.

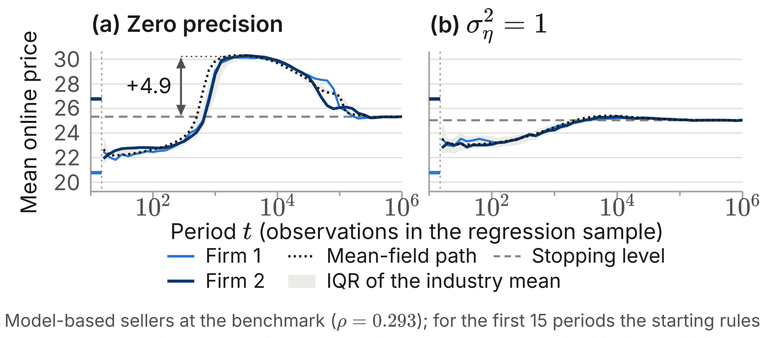
7 Would sellers choose exact quotes?



A model-based seller's best wedge variance against its rival's (benchmark, truncated cost moments; legend values are nominal, and 0.3 becomes 0.293 after truncation).

- Against an uninformed rival, full precision is best (2.7% more profit than zero precision at $\rho = 0.293$), mostly because the rival's regression credits the seller's response to the rival's own price and the rival prices higher. This holds for rival wedge variances above about 0.6.
- **Proposition 4:** against a fully precise rival, a small loss of precision pays if and only if the mean squared Nash online margin exceeds a threshold proportional to cost dispersion; then full precision for both is not an equilibrium, even when free. At the benchmark the best partial precision earns about 0.25% more, pricing as if the rival followed with weight 0.247, near the actual 0.25.
- Precision choices are strategic substitutes. Alternating best responses reach asymmetric equilibria with mean prices 1.0% to 4.3% above the Nash mean (ρ from 0 to 0.3); moving both sellers to full precision raises consumer surplus by 2.6% to 2.8% and commission revenue.
- A seller that can change its rule, not only its precision, keeps exact quotes and acts on the rival's measured response; if both sellers do so, the mean price is about 6% above the Nash mean (paper Remark 2).
- With a correct belief, rule-tuning sellers have full precision as a dominant strategy, worth less than 0.02% of profit at the benchmark.

8 Price overshooting during learning

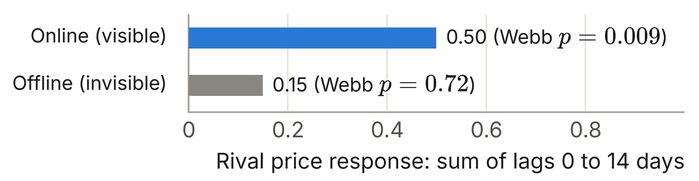


Model-based sellers at the benchmark ($\rho = 0.293$); for the first 15 periods the starting rules have mean prices 3 below and 3 above the Nash mean (marks at left). Medians of 20 runs with forgetting window $W = 5 \times 10^4$ (the regression's effective memory); +4.9 is the peak of the median industry mean above the stopping level.

From this start, zero-precision prices fall below the Nash mean and stay there for about 300 periods, then rise together to 4.9 above the stopping level: the regression credits the shared drift to the own price. A moderately precise quote ($\sigma_\eta^2 = 1$) cuts the overshoot to 0.23.

9 Pharmacy panel data

One city · 6 trade areas · 3,937 SKUs · 6 chain and 14 rival outlets · 6 Jan to 6 Aug 2026



Full sample: sums over lags 0 to 14 days of the coefficients on the chain's daily price changes in each channel, in regressions of rival online price changes dated at the last day between two records (at the midpoint or first day, the online sum is small and insignificant). Their difference has Webb $p = 0.21$ (wild cluster bootstrap by trade area).

- Rival online prices move with the chain's visible online price changes and not significantly with its invisible offline changes. Aligned to the rivals' recording intervals (a different estimand), the difference between the two channels, 0.048, is significant (Webb $p = 0.040$).
- Offline prices are sticky: 82% of the chain's product-store pairs never change the offline price, against 60% online. Online, the chain reprices by a fixed judgment rule, with a median new price 0.3% below the cheapest rival; its management says rivals also reprice by judgment.
- Around its own online price changes, the chain's online units rise after cuts and fall after increases, with a two-week response of -0.41 ± 0.27 log units per unit log price change (Webb $p = 0.054$), as the own-price term of the demand system predicts.

Conclusion

Precision moves long-run mean prices only where sellers re-estimate demand, and there a small improvement from low precision can raise them. Even when precision is free, such sellers need not both choose exact quotes: at the benchmark they do not, and their equilibrium mean prices are 1.0% to 4.3% above the Nash mean. Sellers that act on the rival's measured response keep exact quotes but also price above the Nash mean.

What precision does for each type of seller

Seller	Conjecture Λ	Mean price, zero to full precision	First improvement from zero precision	Consumer surplus, zero to full precision
● **Model-based (M)**	ρ at zero, 0 at full precision (Thm. 1)	falls if $\rho > 0$	raises the mean price if and only if $\rho < \rho_{\mathrm{nm}}(\omega)$ (Thm. 2)	rises if and only if $\rho > \rho^\star$ (Prop. 3)
○ **Rule-tuning, correct belief (R)**	0 (Thm. 3)	unchanged at the Nash mean	no effect on the mean	falls (Prop. 2)
○ **Rule-tuning, belief $\tilde{b}$**	$(b - \tilde{b})/e$ (Thm. 3)	unchanged at $\bar{P}(\tilde{b})$	no effect on the mean	either sign (Thm. 3)
● **Fixed rule (F)**	none: the rule is never revised	unchanged	no effect on the mean	does not rise (Prop. 5)

Each entry holds under the conditions of the cited result: Proposition 3 requires condition (23), which holds at the benchmark; the consumer-surplus entries for R and F require $\omega \le 1/2$. At the benchmark $\rho_{\mathrm{nm}}(0.25) = 0.268$ and $\rho^\star = 0.00065$. Theorem and proposition numbers follow the paper.

Bresnahan TF (1981) Duopoly models with consistent conjectures. *American Economic Review* 71(5):934–945.
Fudenberg D, Levine DK (1993) Self-confirming equilibrium. *Econometrica* 61(3):523–545.
Light B, Wang W (2026) Conjectural variations in competitive dynamic pricing: A learning foundation via experimentation design and feedback structure. arXiv:2602.12888.

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Data from a research collaboration with a pharmacy retailer; not publicly available.